lit2026 W3: IC and precompute configurations against generic lower bounds

Run `run-20261011T2012Z` · **commit** `3c239129d + examples/lit2026_w3_ggm_audit.rs (this PR)` · **host** Apple M4 Pro, 14 cores, 48 GiB, macOS 26.6, shared (load 80-91 during the run), rustc 1.93.1; isolation L0 · **date** 2026-10-11 (UTC start 20:12). Papers: Corrigan-Gibbs–Henzinger–Wu, *The Structured Generic-Group Model*, ePrint 2026/384 (EUROCRYPT 2026); Dinur–Keller–Marmor, arXiv 2505.00894 (STOC 2026); Akshima–Kiltz–Nishaburi–Nooripoor–Wiedijk, ePrint 2026/2253 (TCC 2026). Producer: `examples/lit2026_w3_ggm_audit.rs` (`table`, `fit`, `measure`). Plot: `plot.py` (layout only). Scope: public synthetic toy curves; no key-recovery, asymptotic or deployed-curve claim; multi-target results are secondary diagnostics.

Bounds used

- **SGGM, Theorem 3.2.** Prime order r; δ = fraction of labels constrained by $\star$ (Def. 3.1: inputs or output of a defined $\star$). For every T-query algorithm, $\mathrm{Adv} \le \delta(3T+2) + \frac{(3T+1)^2}{r} + \frac{1}{r}$, so constant success needs $T = \Omega\big(\min\big(\sqrt{r}, \frac{1}{\delta}\big)\big)$ group-operation queries; $\star$ evaluations are free. Appendix B (Thm B.1): with S advice bits, $\mathrm{Adv} \le \tilde{O}\big(S\frac{T^2}{r} + \delta T\big)$.
- **Mapping to IC.** Labels = points of the order-r subgroup. An m-term decomposition oracle over factor base F is a partial $\star$ defined on pairs of j-sums ($j \le m$) whose sum is an $(i+j)$-sum; the constrained set is F and every j-sum of F, so $\delta \le c\sum_{j=1}^{m}\binom{|F|+j-1}{j}/r$ (c = torsion classes a relation may absorb; we take $c = h$). We use $\delta = \max(\text{that count}, \text{measured relations/trials})$, the weakest bound any record admits. The paper's copy we read (UT Austin PDF of the EUROCRYPT version, 2026-02-24) has no elliptic-curve section; the abstract's EC claim could not be checked because eprint.iacr.org sits behind a bot challenge.
- **Automorphisms.** Thm 3.2 has none. We report $\min\big(\sqrt{r/A}, 1/\delta\big)$ as asked and $\min\big(\sqrt{r/A}, 1/(A\delta)\big)$ (each query read as A labels; our extension, the weaker).
- **DKM Theorem 1.1.** Non-adaptive (S, T) (pre-translated queries fixed by the advice, not by the challenge): $\mathrm{Adv} \le 3T^2/N + 4\ln 2 \cdot ST/N$, so $ST \gtrsim N$ below $T = \sqrt{N}$. Adaptive algorithms reach $ST^2 \approx N$ (Corrigan-Gibbs–Kogan; Bernstein–Lange).
- **Multi-instance.** k generic logarithms cost $\Theta\big(\sqrt{kr}\big)$ in total (Yun 2015; 2026/2253 abstract: $\frac{1}{2}(\log p + \log k)$ bits for multi-instance CDH/SDH), so $d\ln(\text{total})/d\ln k \to 1/2$; Kuhn–Struik's law gives the finite-k slope.

Result 1: no measured cost is below its bound

family	rows	$\delta_{\text{upper}} < 1$	min measured / bound	min measured / bound, $\delta_{\text{upper}} < 1$	rows below a bound
prime	72	48	38.868	38.868	0
char2	35	23	56.777	56.777	0
koblitz_round5	72	13	355	616	0
koblitz	6	1	2.84e6	2.84e6	0
koblitz_batch	9	1	79898	1.96e6	0
koblitz_compact_orbit	14	0	1077	—	0

Units: GAE = group-addition equivalents, whole process ($S \cdot \sqrt{r} \cdot k$, S from `docs/ic/leaderboard.json` and `docs/ic/boundary_targets.json`); compact-orbit rows are probes (S3 solve + root-table lookup), not group operations. Of 208 rows, 122 have $\delta_{\text{upper}} = 1$: their factor base's m-sums cover the subgroup, so the bound is $\min\big(\sqrt{r/A}, 1\big) = 1$ and says nothing.

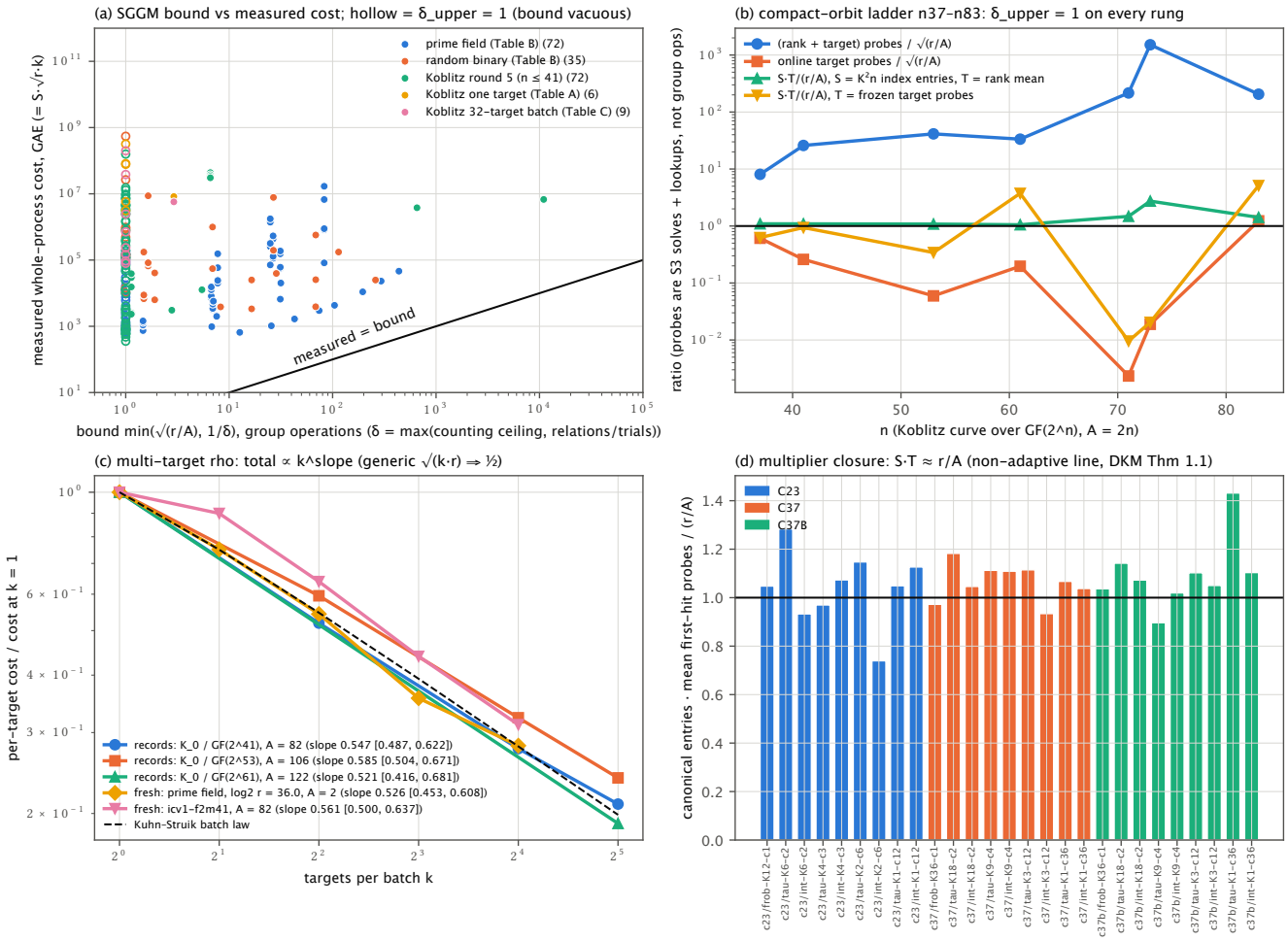


Figure 1: Recorded values from run `run-20261011T2012Z`. (a) every leaderboard/round-5 configuration, bound vs measured GAE; hollow = vacuous ($\delta_{\text{upper}} = 1$). (b) compact-orbit ladder ratios. (c) per-target multi-target rho cost relative to $k = 1$, committed records (16/16/8 batches) and the fresh run (24 batches per k). (d) multiplier closure: canonical entries × mean first-hit probes over r/A.

Rows where the bound is not vacuous ($\delta_{\text{upper}} < 1$)

configuration	log2 r	A	m	|F|	δ meas.	δ_{upper}	bound	measured GAE	ratio
icv1-fp10-t5-192cb216/mitm_m2_negfold_walk_balanced	9.68	2	2	10	0.062	0.079	12.662	655	51.771
icv1-fp10-t5-192cb216/mitm_m2_negfold_walk	9.68	2	2	32	0.417	0.68	1.47	744	506
icv1-fp10-t5-192cb216/mitm_m2_negfold	9.68	2	2	32	0.467	0.68	1.47	1054	717
icv1-fp10-t5-192cb216/semaev_s3_roots_m2	9.68	2	2	32	0.467	0.68	1.47	1095	745
icv1-fp10-t5-192cb216/mitm_m2	9.68	2	2	32	0.467	0.68	1.47	1310	891
icv1-fp10-t5-192cb216/direct_subtraction_m2	9.68	2	2	32	0.467	0.68	1.47	1451	987
icv1-fp12-t19-d4a315df/mitm_m2_negfold_walk	11.93	2	2	32	0.147	0.144	6.786	974	144
icv1-fp12-t19-d4a315df/mitm_m2_negfold_walk_balanced	11.93	2	2	16	0.03	0.039	25.586	1033	40.392
icv1-fp12-t19-d4a315df/mitm_m2_negfold	11.93	2	2	32	0.127	0.144	6.945	3217	463
icv1-fp12-t19-d4a315df/mitm_m2	11.93	2	2	32	0.127	0.144	6.945	3473	500
icv1-fp12-t19-d4a315df/semaev_s3_roots_m2	11.93	2	2	32	0.127	0.144	6.945	4676	673
icv1-fp12-t19-d4a315df/direct_subtraction_m2	11.93	2	2	32	0.127	0.144	6.945	5701	821
icv1-fp14-tm43-158b6914/mitm_m2_negfold_walk_balanced	13.99	2	2	26	0.023	0.023	42.857	1666	38.868
icv1-fp14-tm43-158b6914/mitm_m2_negfold_walk	13.99	2	2	64	0.132	0.132	7.57	1998	264
icv1-fp14-tm43-158b6914/mitm_m2_negfold	13.99	2	2	64	0.149	0.132	6.722	7150	1064
icv1-fp14-tm43-158b6914/mitm_m2	13.99	2	2	64	0.149	0.132	6.722	8174	1216
icv1-fp14-tm43-158b6914/semaev_s3_roots_m2	13.99	2	2	64	0.149	0.132	6.722	12687	1887
icv1-fp14-tm43-158b6914/direct_subtraction_m2	13.99	2	2	64	0.149	0.132	6.722	15424	2295
icv1-fp16-t351-d433f24e/mitm_m2_negfold_walk_balanced	15.99	2	2	40	0.013	0.013	74.102	2979	40.198

configuration	log2 r	A	m	|F|	δ meas.	δ_{upper}	bound	measured GAE	ratio
icv1-fp16-t351-d433f24e/mitm_m2_negfold_walk	15.99	2	2	128	0.142	0.129	7.037	5736	815
icv1-fp16-t351-d433f24e/mitm_m2_negfold	15.99	2	2	128	0.117	0.129	7.753	20120	2595
icv1-fp16-t351-d433f24e/mitm_m2	15.99	2	2	128	0.117	0.129	7.753	24216	3123
icv1-fp16-t351-d433f24e/direct_subtraction_m2	15.99	2	2	128	0.117	0.129	7.753	58808	7585
icv1-fp16-t351-d433f24e/semaev_s3_roots_m2	15.99	2	2	128	0.117	0.129	7.753	154746	19959
icv1-fp18-t175-45770ebc/mitm_m2_negfold_walk_balanced	18	2	2	64	0.01	0.008	104	4304	41.233
icv1-fp18-t175-45770ebc/mitm_m2_negfold_walk	18	2	2	128	0.03	0.032	31.211	6601	212
icv1-fp18-t175-45770ebc/mitm_m2_negfold	18	2	2	128	0.031	0.032	31.211	56718	1817
icv1-fp18-t175-45770ebc/mitm_m2	18	2	2	128	0.031	0.032	31.211	60814	1948
icv1-fp18-t175-45770ebc/semaev_s3_roots_m2	18	2	2	128	0.031	0.032	31.211	154578	4953
icv1-fp18-t175-45770ebc/direct_subtraction_m2	18	2	2	128	0.031	0.032	31.211	187797	6017
icv1-fp20-t1293-3c2999ba/mitm_m2_negfold_walk_balanced	20	2	2	102	0.005	0.005	196	10972	56.117
icv1-fp20-t1293-3c2999ba/mitm_m2_negfold_walk	20	2	2	256	0.031	0.032	31.582	20259	641
icv1-fp20-t1293-3c2999ba/mitm_m2_negfold	20	2	2	256	0.038	0.032	26.589	112963	4249
icv1-fp20-t1293-3c2999ba/mitm_m2	20	2	2	256	0.038	0.032	26.589	129347	4865
icv1-fp20-t1293-3c2999ba/semaev_s3_roots_m2	20	2	2	256	0.038	0.032	26.589	434803	16353
icv1-fp20-t1293-3c2999ba/direct_subtraction_m2	20	2	2	256	0.038	0.032	26.589	536937	20194
icv1-fp22-tm1267-a5acaf6d/mitm_m2_negfold_walk_balanced	21.65	2	2	148	0.003	0.003	295	23169	78.65
icv1-fp22-tm1267-a5acaf6d/mitm_m2_negfold_walk	21.65	2	2	512	0.037	0.04	24.967	71270	2855
icv1-fp22-tm1267-a5acaf6d/mitm_m2_negfold	21.65	2	2	512	0.04	0.04	24.967	253057	10136
icv1-fp22-tm1267-a5acaf6d/mitm_m2	21.65	2	2	512	0.04	0.04	24.967	318593	12760
icv1-fp22-tm1267-a5acaf6d/semaev_s3_roots_m2	21.65	2	2	512	0.04	0.04	24.967	1.39e6	55772
icv1-fp22-tm1267-a5acaf6d/direct_subtraction_m2	21.65	2	2	512	0.04	0.04	24.967	1.76e6	70513
icv1-fp24-t1577-77336155/mitm_m2_negfold_walk_balanced	23.38	2	2	222	0.002	0.002	438	46179	105
icv1-fp24-t1577-77336155/mitm_m2_negfold_walk	23.38	2	2	512	0.012	0.012	82.932	82157	991
icv1-fp24-t1577-77336155/mitm_m2_negfold	23.38	2	2	512	0.012	0.012	82.932	821584	9907
icv1-fp24-t1577-77336155/mitm_m2	23.38	2	2	512	0.012	0.012	82.932	887120	10697
icv1-fp24-t1577-77336155/direct_subtraction_m2	23.38	2	2	512	0.012	0.012	82.932	6.73e6	81201
icv1-fp24-t1577-77336155/semaev_s3_roots_m2	23.38	2	2	512	0.012	0.012	82.932	1.69e7	203701
icv1-f2m15-t131-ed7256f0/mitm_m2_negfold_walk	13.99	2	2	30	0.018	0.061	16.484	3361	204
icv1-f2m15-t131-ed7256f0/mitm_m2_negfold_walk_balanced	13.99	2	2	14	0.003	0.015	68.567	3893	56.777
icv1-f2m15-t131-ed7256f0/mitm_m3_negfold_walk	13.99	2	3	30	0.1	0.669	1.496	6785	4536
icv1-f2m15-t131-ed7256f0/mitm_m3_negfold	13.99	2	3	30	0.144	0.669	1.496	8587	5741
icv1-f2m15-t131-ed7256f0/mitm_m3	13.99	2	3	30	0.144	0.669	1.496	8812	5891
icv1-f2m15-t131-ed7256f0/mitm_m2_negfold	13.99	2	2	30	0.013	0.061	16.484	25071	1521
icv1-f2m15-t131-ed7256f0/semaev_s4_pairs_and_solve_m3	13.99	2	3	30	0.134	0.669	1.496	174243	116490
icv1-f2m18-t393-d2c04152/mitm_m2_negfold_walk_balanced	15	2	2	30	0.004	0.121	8.262	3854	466
icv1-f2m18-t393-d2c04152/mitm_m2_negfold_walk	15	2	2	64	0.017	0.524	1.908	6308	3307
icv1-f2m18-t393-d2c04152/mitm_m2_negfold	15	2	2	64	0.016	0.524	1.908	40875	21428
icv1-f2m21-t2591-de54d72e/mitm_m2_negfold_walk_balanced	20	2	2	62	0.001	0.004	260	24961	96.05
icv1-f2m21-t2591-de54d72e/mitm_m2_negfold_walk	20	2	2	122	0.003	0.015	68.674	25530	372
icv1-f2m21-t2591-de54d72e/mitm_m3_negfold_walk	20	2	3	122	0.122	0.607	1.648	66352	40263
icv1-f2m21-t2591-de54d72e/mitm_m3_negfold	20	2	3	122	0.142	0.607	1.648	79156	48032
icv1-f2m21-t2591-de54d72e/mitm_m3	20	2	3	122	0.142	0.607	1.648	82877	50290
icv1-f2m21-t2591-de54d72e/mitm_m2_negfold	20	2	2	122	0.004	0.015	68.674	570242	8304
icv1-f2m21-t2591-de54d72e/semaev_s4_pairs_and_solve_m3	20	2	3	122	0.141	0.607	1.648	8.72e6	5.29e6
icv1-f2m24-tm3047-2d0c0e8c/mitm_m2_negfold_walk_balanced	21	2	2	134	0.001	0.035	28.564	39452	1381
icv1-f2m24-tm3047-2d0c0e8c/mitm_m2_negfold_walk	21	2	2	274	0.005	0.145	6.909	55081	7972
icv1-f2m24-tm3047-2d0c0e8c/mitm_m2_negfold	21	2	2	274	0.004	0.145	6.909	993140	143745
icv1-f2m27-t12543-569dca8b/mitm_m2_negfold_walk_balanced	24.41	2	2	254	2.58e−4	0.009	114	174655	1529

configuration	log2 r	A	m	|F|	δ meas.	δ_{upper}	bound	measured GAE	ratio
icv1-f2m27-t12543-569dca8b/mitm_m2_negfold_walk	24.41	2	2	526	0.001	0.037	26.795	197196	7359
icv1-f2m27-t12543-569dca8b/mitm_m2_negfold	24.41	2	2	526	0.001	0.037	26.795	7.78e6	290286
icv1-f2m41-tm2308219-7f48b14a/koblitz_one_target	39	82	3	6560	—	0.343	2.918	8.28e6	2.84e6
icv1-f2m41-tm2308219-7f48b14a/koblitz_collection_m3_aimed	39	82	3	6560	0.009	0.343	2.918	5.71e6	1.96e6
K_1 / GF(2^17)/mitm_m2_signed_orbit_columns	16	34	2	239	—	0.883	1.133	29594	26117
K_1 / GF(2^17)/mitm_m2_abscissa_columns_control	16	34	2	239	—	0.883	1.133	38733	34183
K_1 / GF(2^17)/mitm_m2_signed_orbit_columns_negfold	16	34	2	239	—	0.883	1.133	15444	13630
K_1 / GF(2^17)/mitm_m2_signed_orbit_columns_frobfold	16	34	2	239	—	0.883	1.133	2524	2227
K_1 / GF(2^17)/mitm_m2_signed_orbit_columns_frobfold_walk	16	34	2	239	—	0.883	1.133	2323	2050
K_1 / GF(2^19)/mitm_m2_signed_orbit_columns_frobfold_walk	18	38	2	305	—	0.358	2.791	3063	1098
K_1 / GF(2^23)/mitm_m2_signed_orbit_columns_frobfold_walk	22	46	2	875	—	0.183	5.46	12685	2324
K_0 / GF(2^41)/mitm_m3_signed_orbit_columns	39	82	3	5003	—	0.152	6.577	4.36e7	6.63e6
K_0 / GF(2^41)/mitm_m3_signed_orbit_columns_negfold	39	82	3	5003	—	0.152	6.577	3.74e7	5.68e6
K_0 / GF(2^41)/mitm_m3_signed_orbit_columns_frobfold	39	82	3	5003	—	0.152	6.577	3.35e7	5.09e6
K_0 / GF(2^41)/mitm_m3_signed_orbit_columns_frobfold_walk	39	82	3	5003	—	0.152	6.577	3.04e7	4.63e6
K_0 / GF(2^41)/mitm_m2_signed_orbit_columns_frobfold_walk	39	82	2	5003	—	9.11e−5	10975	6.76e6	616
K_0 / GF(2^41)/mitm_m2_signed_orbit_columns_frobfold_walk_balanced	39	82	2	20501	—	0.002	654	3.79e6	5800

Measured relation rate above the counting ceiling
Relations per trial is a mean number of representations; for uniform targets its expectation is at most the count. 13 rows exceed it; tested as a Poisson excess over all trials (one-sided, Bonferroni level 0.05/100):

row	observed relations	expected at ceiling	z	p
icv1-fp12-t19-d4a315df/mitm_m2_negfold_walk	42	41.039	0.072	0.471
icv1-fp14-tm43-158b6914/mitm_m2_negfold_walk_balanced	35	34.841	−0.058	0.523
icv1-fp14-tm43-158b6914/mitm_m2_negfold	72	63.933	0.946	0.172
icv1-fp14-tm43-158b6914/mitm_m2	72	63.933	0.946	0.172
icv1-fp14-tm43-158b6914/semaev_s3_roots_m2	72	63.933	0.946	0.172
icv1-fp14-tm43-158b6914/direct_subtraction_m2	72	63.933	0.946	0.172
icv1-fp16-t351-d433f24e/mitm_m2_negfold_walk_balanced	49	48.039	0.067	0.473
icv1-fp16-t351-d433f24e/mitm_m2_negfold_walk	134	122	1.077	0.141
icv1-fp18-t175-45770ebc/mitm_m2_negfold_walk_balanced	55	47.038	1.088	0.138
icv1-fp20-t1293-3c2999ba/mitm_m2_negfold	197	166	2.38	0.009
icv1-fp20-t1293-3c2999ba/mitm_m2	197	166	2.38	0.009
icv1-fp20-t1293-3c2999ba/semaev_s3_roots_m2	197	166	2.38	0.009
icv1-fp20-t1293-3c2999ba/direct_subtraction_m2	197	166	2.38	0.009

None is significant after correction; the smallest p is the shared `fp20` random-target trial stream (four variants report the same trials).

Result 2: the compact-orbit ladder (n37–n83)

rung	log2 r	B	K	δ_{upper}	target probes	over $\sqrt{r/A}$	log2(S·T²) − log2(r/A)	S·T/(r/A) frozen / rank mean	frozen / rank mean (P)
compact-n37	27.78	518	7	1	1077	0.61	9.4	0.627 / 1.094	0.573 (0.436)
compact-n41	39	6970	85	1	21248	0.26	14.3	0.939 / 1.092	0.86 (0.577)
compact-n53	44.26	23320	220	1	26485	0.059	13.1	0.342 / 1.081	0.316 (0.271)
compact-n61	47.21	73200	600	1	227437	0.197	19.7	3.741 / 1.053	3.554 (0.971)
compact-n71	52.29	85200	600	1	14554	0.002	7.1	0.01 / 1.478	0.006 (0.006)
compact-n73	56.26	87600	600	1	457561	0.019	13.2	0.02 / 2.734	0.007 (0.007)
compact-n83	52.93	99600	600	1	8.85e6	1.231	25.4	5.12 / 1.417	3.613 (0.973)

$S = K^2 n$ regular states of the S3 root index (advice); T = probes. $\delta_{\text{upper}} = 1$ on every rung ($B^4/24 > r$), so the SGGM bound is vacuous: the cost lies in finding a decomposition, which the model treats as a free $\star$ evaluation. Against preprocessing lines the online phase is consistent: $\log_2(ST^2) - \log_2(r/A)$ is 7.1 to 25.4 with the frozen target's probes, and at the rank-stage mean probe count $ST/(r/A)$ is 1.053 to 2.734, i.e. the target scan sits on the **non-adaptive** line $ST \approx N$.

Result 3: precompute arms and adaptivity

Multiplier closure (27 arms, cells C23/C37/C37b): canonical entries × mean first-hit probes / (r/A) ranges 0.739 – 1.432 pairs × probes / r ranges 0.913 – 2.251. Each arm below is classified by whether its online queries depend on earlier answers (adaptive) or are fixed by the advice and the challenge alone (non-adaptive, DKM's sense: pre-translated queries independent of the challenge).

arm	online phase	class	what the bound implies
S3 root Bloom lookup (PR #1659, `KIC_ROOT_BLOOM=1`)	Same probe list as the exact root table (fixed state order, keyed by the target's x); the filter only skips exact lookups that would miss; outputs and the 71-probe count are identical in both modes (N23 receipt).	non-adaptive table scan; Bloom adds 8 bits per root entry to S, changes no query	Moves no configuration relative to any bound: S grows by a constant factor, T (probes) is unchanged. At most a constant per probe; it cannot lower the S·T line it sits on.
Multiplier closure (Lead A1)	Random (c, d) drawn independently of R and of earlier answers; probe T = R – s_c – s_d against the pair table; stop at the first hit.	non-adaptive	DKM Thm 1.1: S·T ≳ N below T = √N. Measured canonical·T/(r/A) = 0.739 to 1.432 on all 27 arms: the arm sits on that line and cannot go below it without adaptivity. Matches the earlier "precompute-only" finding (PR #1648).
Compact-orbit target extraction (n37–n83)	Fixed state × shift order; S3 roots of the target's x looked up in the K^2n-state index; stop at the first lift.	non-adaptive scan	At the rank-stage mean, S·T/(r/A) = 1.053 to 2.734: on the non-adaptive line. The frozen n71 and n73 targets sit at S·T/(r/A) = 0.01 and 0.02, one draw each, with P(T ≤ frozen) = 0.006 and 0.007 under an exponential first-hit law at the rank-stage mean.
Pair-root tables / self-collision (Lead B)	None: relations come from sorting the stored pair states; no target.	precompute only (no online queries)	Outside DKM (which bounds online T). Changes the rank-stage precompute only; no measurement exists.
Preprocessing rho (Bernstein–Lange; `KIC_RHO_PRECOMPUTE_WALKS`, `preprocessing_rho.rs`)	Walk from the target; each step depends on the previous point.	adaptive	Not bound by DKM; Corrigan-Gibbs–Kogan S· T^2 ≈ N is tight. At equal S it needs T ≈ √(N/S), against N/S for any non-adaptive arm. Records are wall-time only (n53, 1,024 targets), so no ops placement here.

Result 4: multi-target slope

source	curve	A	k	slope	95% bootstrap	Kuhn–Struik slope
records	K_0 / GF(2^41)	82	1,4,16,32	0.547	[0.487, 0.622]	0.533
records	K_0 / GF(2^53)	106	1,4,16,32	0.585	[0.504, 0.671]	0.533
records	K_0 / GF(2^61)	122	1,32	0.521	[0.416, 0.681]	0.534
fresh	prime-floor q=68723001073 N=68723001023 (ell=120283, t=51)	2	1,2,4,8,16	0.526	[0.453, 0.608]	0.539
fresh	icv1-f2m41-tm2308219-7f48b14a	82	1,2,4,8,16	0.561	[0.5, 0.637]	0.539

Slope = OLS of ln(mean batch total GAE) on ln k; interval = 2,000 bootstrap resamples of batches within each k. Fresh run: all logarithms verified against the planted ones: true, true.

Record checks

- `docs/ic/boundary_targets.json /regimes/koblitz/records/vs_rho/history/2/metrics/subgroup_log2_order`: recorded log2 r = 47.5407 log2 of the exact r = 47.2109.
- `docs/ic/boundary_targets.json /regimes/koblitz/records/vs_rho/history/2/metrics/ic_precomputation_ms_excluded_from_online` (n = 61): recorded 13098 ms excluded; the rung's own components (`experiments/koblitz-single-target-n61-20260925/claim_report_vs_rho.json ic_precompute_detail|precompute_detail`): base load 653, index build 12441, rank stage 11053, linear algebra 0.781 ms. The recorded figure equals base_load + index_build + linear_algebra (relative error 1.74e−4); it leaves out the rank stage.
- `docs/ic/boundary_targets.json /regimes/koblitz/records/vs_rho/history/1/metrics/ic_precomputation_ms_excluded_from_online` (n = 71): recorded 39113 ms excluded; the rung's own components (`experiments/koblitz-single-target-n71-20261002/claim_report_vs_rho.json ic_precompute_detail|precompute_detail`): base load 290, index build 38807, rank stage 1.16e6, linear algebra — ms. The recorded figure equals base_load + index_build (relative error 4.1e−4); it leaves out the rank stage.